

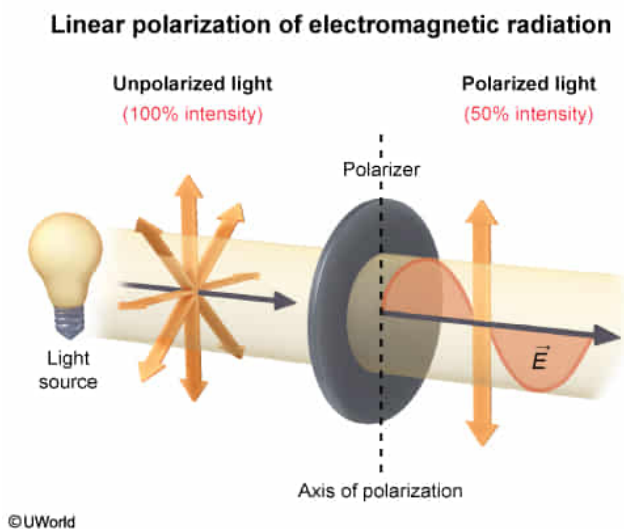
Light from an incandescent lightbulb passes through an ideal linear polarizer. The intensity of the light is 100 lm before passing through the filter, but 50 lm immediately after. Which of the following explains this discrepancy?

- ☐ A. Refraction occurs as light enters the polarizer. [18%]
- ☐ B. Light waves with magnetic fields oriented perpendicular to the axis of polarization are absorbed. [22%]
- ✓ ☒ C. Light waves with electric fields oriented perpendicular to the axis of polarization are absorbed. [51%]
- ☐ D. Reflection occurs before light enters the polarizer. [8%]

Correct

51% answered correctly

Explanation:



Polarization refers to the alignment of transverse wave **oscillations** in a particular orientation within the x,y,z -coordinate system. For example, a transverse wave traveling along the x -axis causes **oscillations** that may occur exclusively along the y - or z -axis, or at some angle between the y - and z -axes. Polarization is unique to **transverse waves** because only transverse waves cause oscillations perpendicular to the direction of propagation. Consequently, electromagnetic radiation can be polarized but sound cannot.

A waveform is said to be **linearly polarized** if waveform oscillations take place within only one plane of the coordinate system. By standard convention, the overall orientation of electromagnetic radiation is the same as the orientation of the electric field component.

Polarization filters are optical devices that reorient the polarization of electromagnetic radiation so that radiation exiting the filter is polarized in only one orientation. A **linear polarization filter**, for example, allows for the transmission of electromagnetic radiation oriented parallel to the **axis of polarization** but inhibits the passage of radiation oriented perpendicular to this axis.

Most light sources emit unpolarized light comprising waveforms oriented in all directions. Even for unpolarized light, however, approximately half of the intensity of electromagnetic radiation is polarized along one of two perpendicular axes. Because only electromagnetic radiation with an electric field oriented parallel to the axis of polarization passes through, the total intensity of light will therefore decrease by 50% when nonpolarized light passes through a linear polarization filter.

(Choices A and D) The reflection and refraction of light are important to an optical phenomenon related to a discrepancy between the **refractive index** of two media. However, polarizers are designed to facilitate the passage of light unrefracted and unreflected.

(Choice B) Light waves with magnetic fields oriented perpendicular to the axis of polarization have electric fields parallel to the axis of polarization. Such waves are transmitted through the polarizer and are not absorbed.

Educational objective:

Linearly polarized waveforms are associated with oscillations that occur in a particular orientation of the x,y,z-coordinate system. A linear polarization filter blocks electric fields aligned perpendicular to the axis of polarization (ie, 50% of total intensity).

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Subject:
Physics

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System:
Light and Sound